

BISHKEK, Kyrgyzstan

WMO: 383530

Lat: **42.850N**Lon: **74.533E**Elev: **2493**StdP: **13.42**Time Zone: **6.00 (E06)**Period: **97-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	1.3	7.2	-3.3	5.1	2.8	1.7	6.6	8.8	16.3	38.1	12.7	38.1	2.3	180	0.421

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	22.8	95.9	65.7	93.2	64.5	90.7	63.8	70.5	89.1	67.8	86.0	65.8	84.6	5.4	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
64.0	98.0	80.8	61.0	88.0	76.3	59.0	81.6	74.9	36.1	88.6	33.7	85.9	32.0	84.6	82.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4)	16.5	13.2	10.4	DB	-2.3	101.1	7.8	2.9	-7.9	103.1	-12.4	104.8	-16.8	106.4	-22.5	108.5
(5)				WB	-2.9	70.4	7.6	5.1	-8.3	74.1	-12.8	77.1	-17.0	79.9	-22.5	83.7

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	53.9	28.7	32.3	45.1	55.5	64.8	73.5	77.6	75.9	66.1	53.8	41.3	30.7
	DBStd	19.20	10.14	10.42	9.98	8.49	6.73	6.01	5.20	5.49	7.47	8.73	9.74	10.24
	HDD50	2365	661	498	212	43	2	0	0	0	1	61	290	597
	HDD65	5262	1125	916	617	297	86	9	0	2	77	358	713	1062
	CDD50	3787	0	3	61	208	460	705	855	802	486	178	28	1
	CDD65	1210	0	0	1	12	80	264	391	339	111	11	1	0
	CDH74	13740	0	0	22	185	836	2887	4619	3870	1176	134	11	0
	CDH80	5989	0	0	3	36	239	1232	2250	1789	408	31	1	0
Wind	WSAvg	3.9	3.1	3.6	4.2	4.6	4.5	4.6	4.4	4.2	3.8	3.5	3.2	3.1
Precipitation	PrecAvg	17.3	1.0	1.3	1.8	2.9	2.6	1.4	0.7	0.5	0.7	1.5	1.7	1.2
	PrecMax	26.4	2.3	3.3	4.1	6.2	6.2	3.8	3.4	2.0	2.2	4.0	4.6	3.3
	PrecMin	11.4	0.2	0.3	0.6	0.5	0.9	0.2	0.0	0.0	0.0	0.2	0.3	0.2
	PrecStd	3.3	0.5	0.6	0.8	1.3	1.3	0.9	0.8	0.4	0.6	0.9	0.9	0.7
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	56.9	63.2	77.3	84.3	91.1	98.6	100.4	98.2	92.6	84.4	74.4	57.9
		MCWB	43.7	47.3	55.5	59.1	67.1	66.9	65.7	65.9	63.4	58.1	53.4	43.4
	2%	DB	50.0	55.5	70.7	78.9	85.6	93.3	96.6	94.9	88.5	77.3	64.9	52.0
		MCWB	39.7	43.9	51.9	57.7	63.2	64.8	65.6	64.8	61.4	55.1	49.5	40.7
	5%	DB	45.4	50.4	65.5	74.9	82.2	90.2	93.5	92.1	84.7	72.9	59.6	47.5
		MCWB	37.0	41.0	50.0	56.2	61.7	64.2	64.5	64.0	60.2	53.2	46.9	38.5
	10%	DB	41.2	46.1	60.7	70.5	78.6	87.1	90.8	89.3	81.0	68.1	54.9	43.4
		MCWB	34.9	38.9	48.3	54.7	60.2	63.0	63.8	63.4	58.7	51.5	44.5	36.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	44.7	48.3	58.0	63.2	71.6	74.8	74.7	73.5	67.6	60.2	55.1	45.4
		MCDB	54.1	60.9	73.2	78.3	88.7	92.5	94.4	92.1	87.6	77.7	68.6	53.9
	2%	WB	40.7	44.7	53.8	60.5	65.6	69.1	69.7	69.2	63.8	56.4	51.8	41.7
		MCDB	48.7	53.5	65.6	74.7	81.2	86.5	87.5	88.7	83.5	72.7	64.0	50.2
	5%	WB	37.7	41.9	51.1	58.1	63.1	66.2	67.1	66.7	61.3	54.7	47.8	39.2
		MCDB	44.0	49.1	62.6	70.2	77.8	84.0	85.5	86.4	80.3	69.1	56.7	46.5
	10%	WB	35.3	39.2	48.8	55.9	61.1	64.3	65.3	64.7	59.2	52.9	45.3	36.8
		MCDB	40.4	46.0	59.6	67.6	75.8	82.1	84.8	84.8	78.7	65.6	53.6	42.9
Mean Daily Temperature Range	5% DB	MDBR	16.0	16.0	18.6	19.8	21.4	22.6	22.8	23.4	22.6	20.4	16.8	16.0
		MCDBR	22.0	22.5	26.8	27.7	26.8	27.4	26.2	26.6	27.3	27.2	24.5	22.3
		MCWBR	13.9	13.6	13.4	12.8	12.4	11.1	9.2	10.0	11.4	12.4	12.9	13.7
	5% WB	MCDBR	19.7	21.0	24.4	23.9	24.7	25.8	24.8	25.4	26.4	23.8	21.6	20.0
		MCWBR	13.0	13.3	13.2	12.8	13.1	13.0	11.8	12.5	12.9	11.7	12.3	12.9
Clear-Sky Solar Irradiance	taub		0.335	0.381	0.451	0.482	0.455	0.459	0.451	0.418	0.401	0.419	0.384	0.329
	taud		2.134	2.031	1.908	1.900	2.046	2.082	2.119	2.203	2.187	2.034	2.061	2.144
	Ebn at Noon		243	250	246	250	261	260	261	266	260	233	220	232
	Edh at Noon		32	42	54	59	52	51	49	43	41	42	35	29
All-Sky Solar Radiation	RadAvg		662	962	1273	1574	1930	2120	2140	2003	1591	995	641	537
	RadStd		47	73	108	149	145	115	125	108	83	119	69	42

Historical Trends

	DBAvg	
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